

Question-4.10.24

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Question

Find the equations of the lines through the point of intersection of the line $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ and whose distance from the point $(3, 2)$ is $\frac{7}{5}$.

Solution

Given lines are

$$(1 \quad -1) \begin{pmatrix} x \\ y \end{pmatrix} + 1 = 0, (2 \quad -3) \begin{pmatrix} x \\ y \end{pmatrix} + 5 = 0 \quad (1)$$

$$\begin{pmatrix} 1 & -1 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix} \quad (2)$$

Write as augmented matrix

$$\left(\begin{array}{cc|c} 1 & -1 & -1 \\ 2 & -3 & -5 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & -1 & -3 \end{array} \right) \quad (3)$$

From second row: $-y = -3 \implies y = 3$

Substitute in first row: $x - 3 = -1 \implies x = 2$

Intersection point of given lines is $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

Solution

General line through (x_0, y_0) with normal vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is

$$(a \ b) \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \quad (4)$$

Distance from point $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to line is

$$\text{Distance} = \frac{|(a \ b) \left(\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right)|}{\sqrt{(a \ b) \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \quad (5)$$

$$\frac{|(a \ b) \left(\begin{pmatrix} 3-2 \\ 2-3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right)|}{\sqrt{(a \ b) \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \quad (6)$$

Solution

$$\frac{|a-b|}{\sqrt{a^2+b^2}} = \frac{7}{5}$$

$$\text{Let } t = \frac{a}{b} \ (b \neq 0)$$

$$\frac{|t-1|}{\sqrt{t^2+1}} = \frac{7}{5} \quad (7)$$

$$(8)$$

Squaring on both sides

$$25(t-1)^2 = 49(t^2+1) \quad (9)$$

$$25(t^2 - 2t + 1) = 49t^2 + 49 \quad (10)$$

$$25t^2 - 50t + 25 = 49t^2 + 49 \quad (11)$$

$$24t^2 + 50t + 24 = 0 \quad (12)$$

$$12t^2 + 25t + 12 = 0 \quad (13)$$

$$t = \frac{-25 \pm \sqrt{25^2 - 4 \times 12 \times 12}}{2 \times 12} = \frac{-25 \pm 7}{24} \quad (14)$$

Solution

$$t_1 = -\frac{3}{4}, t_2 = -1$$

Normal vectors ($let b = 1$):

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (15)$$

Equation constants $c = -a(2) - b(3)$

$$c_1 = (-2) \times \left(-\frac{3}{4}\right) - 3 \times 1 = \frac{3}{2} - 3 = -\frac{3}{2} \quad (16)$$

$$c_2 = (-2) \times (-1) - 3 \times 1 = 2 - 3 = -1 \quad (17)$$

required equations of line are

$$\begin{pmatrix} -\frac{3}{4} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - \frac{3}{2} = 0 \quad (18)$$

$$\begin{pmatrix} -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 1 = 0 \quad (19)$$

(20)

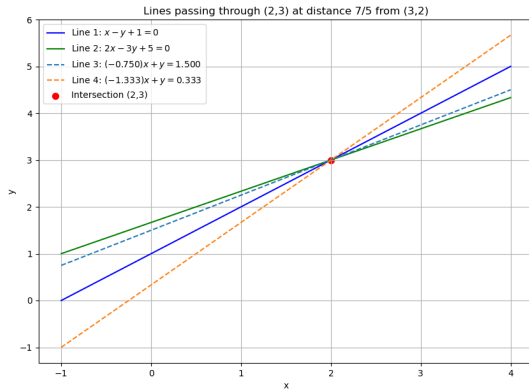


Figure:

```
// code.c
#include <stdio.h>
#include <math.h>

// Solve quadratic  $ax^2 + bx + c = 0$ 
void solve_quadratic(double a, double b, double c, double *r1,
    double *r2) {
    double d = b*b - 4*a*c;
    if (d < 0) {
        *r1 = NAN;
        *r2 = NAN;
        return;
    }
    *r1 = (-b + sqrt(d)) / (2*a);
    *r2 = (-b - sqrt(d)) / (2*a);
}
```



```
void get_line_params(double *a1, double *a2, double *c1, double *
    c2) {
    double A = 1.0, B = 25.0/12.0, C = 1.0;
    double r1, r2;
    solve_quadratic(A, B, C, &r1, &r2);

    a1[0] = r1;
    a2[0] = r2;

    // Constants  $c = 2a + 3$ , intersection point fixed at (2,3)
    c1[0] = 2*r1 + 3;
    c2[0] = 2*r2 + 3;
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
import os  
  
# Load shared library  
lib = ctypes.CDLL(os.path.abspath('./code.so'))
```

```
a1 = ctypes.c_double()
a2 = ctypes.c_double()
c1 = ctypes.c_double()
c2 = ctypes.c_double()

# call C function
lib.get_line_params(ctypes.byref(a1), ctypes.byref(a2), ctypes.
    byref(c1), ctypes.byref(c2))

# Extract values
roots = [a1.value, a2.value]
cs = [c1.value, c2.value]

# Save to numpy for plot
np.save('roots.npy', np.array(roots))
np.save('cs.npy', np.array(cs))
print("Roots (a1, a2):", roots)
print("Constants (c1, c2):", cs)
```

Python-Code

```
import numpy as np
import matplotlib.pyplot as plt

roots = np.load('roots.npy')
cs = np.load('cs.npy')

x = np.linspace(-1, 4, 400)

plt.figure(figsize=(10, 7))

# Original lines
# Line 1:  $x - y + 1 = 0 \Rightarrow y = x + 1$ 
y1 = x + 1
plt.plot(x, y1, label='Line 1:  $x - y + 1=0$ ', color='blue')

# Line 2:  $2x - 3y + 5 = 0 \Rightarrow y = (2x + 5)/3$ 
y2 = (2*x + 5)/3
plt.plot(x, y2, label='Line 2:  $2x - 3y + 5=0$ ', color='green')
```

```
# Required lines (Line 3 and Line 4)
for i in range(2):
    a = roots[i]
    c = cs[i]
    # Equation:  $a x + y = c \Rightarrow y = -a x + c$ 
    y = -a * x + c
    plt.plot(x, y, label=f'Line {i+3}:  $\$({a:.3f})x + y = {c:.3f}\$'$ 
            ', linestyle='--')

# Plot intersection point (2,3)
plt.scatter(2, 3, color='red', s=50, label='Intersection (2,3)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('Lines passing through (2,3) at distance 7/5 from (3,2)')
plt.legend()
plt.grid(True)
plt.axis([-1, 4, -2, 6])
plt.show()
```